

Elena Popescu

Senior NLP Engineer

Senior · Machine Learning Engineering · 9.9 years experience · candidate_024



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Bucharest, Romania

PROFESSIONAL PROFILE

Experienced NLP engineer specializing in production language models, semantic search, and retrieval-augmented applications. Strong hands-on background with Python, PyTorch, vector databases, Dockerized ML services, and model evaluation workflows for multilingual text systems.

PROFESSIONAL EXPERIENCE

Senior NLP Engineer

Apr 2022 - Present

LinguaVector Labs · Bucharest, Romania

- Designed and deployed PyTorch-based NLP pipelines for document understanding, intent classification, and semantic retrieval across Romanian and English datasets.
- Built vector database indexing and retrieval workflows to support low-latency semantic search and retrieval-augmented generation features.
- Containerized training, evaluation, and inference services with Docker, improving reproducibility across development and production environments.
- Partnered with data engineering and product teams to define model quality metrics, offline evaluation sets, and monitoring dashboards for NLP services.

Technologies: Python, PyTorch, NLP, Vector Databases, Docker, FastAPI, MLflow, PostgreSQL

Machine Learning Engineer

Jan 2019 - Mar 2022

NovaText Solutions · Cluj-Napoca, Romania

- Developed text classification, named-entity recognition, and embedding models using Python and PyTorch for customer support and compliance analytics products.
- Implemented semantic similarity services backed by vector databases for duplicate detection and knowledge-base recommendation use cases.
- Optimized model inference APIs and batch scoring jobs, reducing processing time for large text corpora while maintaining evaluation traceability.

Technologies: Python, PyTorch, spaCy, Transformers, Vector Databases, Docker, REST APIs

Data Scientist

Sep 2016 - Dec 2018

Asteria Digital Systems · Bucharest, Romania

- Built early NLP prototypes for sentiment analysis, keyword extraction, and customer feedback clustering using Python and statistical machine learning methods.
- Created reusable preprocessing pipelines for multilingual text normalization, tokenization, and feature extraction.
- Delivered analytical reports and model validation summaries for internal stakeholders and client-facing product teams.

Technologies: Python, scikit-learn, NLP, Pandas, NumPy, SQL, Jupyter

SKILLS

PROGRAMMING LANGUAGES

Python 8.5y

DATA & AI

PyTorch 5.5y

NLP 8y

Transformers 4.5y

Semantic Search 4y

MLflow 3y

scikit-learn 7y

Model Evaluation 6y

DATABASES

Vector Databases 4y

PostgreSQL 5y

DEVOPS & DELIVERY

Docker 5y

FRAMEWORKS

FastAPI 3y

LANGUAGES

Romanian	Native
English	Professional
German	Intermediate

EDUCATION

MSc in Artificial Intelligence

Carpathian Technical University
2014 - 2016 · Bucharest, Romania

BSc in Computer Science

Danube Polytechnic College
2010 - 2014 · Bucharest, Romania

CERTIFICATIONS

Production Machine Learning Professional

Applied AI Institute · 2024